

MeTTa Standard Library Documentation

MeTTa (Meta Type Talk): A Language for Programming with Knowledge

Welcome to the comprehensive documentation for the MeTTa Standard Library!

MeTTa is a cutting-edge, **multi-paradigm programming language** specifically designed for the exciting frontier of **declarative and functional computations over knowledge metagraphs**. It moves beyond traditional programming paradigms, offering a uniquely powerful and flexible environment for building Artificial General Intelligence (AGI) systems and advanced knowledge-based applications.

At its heart, MeTTa embraces a **symbolic approach to AI**, allowing you to represent knowledge, relationships, and reasoning processes in a natural and intuitive way. Unlike languages focused solely on data or algorithms, MeTTa empowers you to **program with meaning**, directly manipulating abstract concepts and complex relationships within its knowledge metagraph, the **AtomSpace**.

Contents:

- Equality and Reduction

- `=` (Equality/Reduction Rule Definition)
- `id` (Identity)
- `=alpha`
- `assertEqual`
- `assertAlphaEqual`
- `assertEqualToResult`
- `assertAlphaEqualToResult`

- Error Handling

- `ErrorType`
- `Error`
- `if-error`
- `return-on-error`

- Evaluation Control

- `return`
- `function`
- `eval`
- `evalc`
- `chain`
- `for-each-in-atom`

- Expression Manipulation

- `cons-atom`
- `decons-atom`
- `car-atom`
- `cdr-atom`
- `size-atom`
- `index-atom`

- Mathematical Operations

- `pow-math`
- `sqrt-math`
- `abs-math`
- `log-math`
- `trunc-math`
- `ceil-math`
- `floor-math`
- `round-math`
- `sin-math`
- `asin-math`
- `cos-math`
- `acos-math`
- `tan-math`
- `atan-math`
- `isnan-math`
- `isinf-math`
- `min-atom`
- `max-atom`
- `random-int`
- `random-float`

- Non-deterministic Computation

- `collapse-bind`
- `superpose-bind`
- `superpose`

- `collapse`

- Type System

- `is-function`
- `type-cast`
- `match-types`
- `first-from-pair`
- `match-type-or`

- List Manipulation

- `filter-atom`
- `map-atom`
- `foldl-atom`
- `format-args`

- Atomspace Interaction

- `add-reduct`
- `add-atom`
- `get-type`
- `get-type-space`
- `get-metatype`
- `if-equal`
- `new-space`
- `new-state`
- `change-state!`
- `get-state`
- `remove-atom`
- `get-atoms`
- `match`

- Quoting

- `quote`
- `unquote`
- `noreduce-eq`

- Set Operations

- `unique`
- `union`
- `intersection`
- `subtraction`
- `unique-atom`



- `union-atom`
- `intersection-atom`
- `subtraction-atom`